Software Engineering Approaches for AI-Enabled Physical Assessment and Personalized Recovery: The BodyBalance AI Framework

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***Abstract*—Artificial intelligence is increasingly applied to physical assessment, exercise guidance, and digital health decision support. Yet posture analysis, pain recording, and activity recommendations are often developed as separate functions, making it difficult to account for a user's overall condition. This paper presents BodyBalance AI, a proposed non-diagnostic mobile decision-support framework integrating daily physical and mental check-ins, a structured Pain Locator, user history, and optional computer-vision-based posture assessment. Drawing on research in pose estimation, digital pain representation, personalized recommendations, and clinical safety, the paper defines a modular architecture with distinct assessment, safety-screening, recommendation, and explanation stages. The design routes predefined concerning indicators away from routine activity guidance and toward professional-evaluation advice. Six predefined decision-support categories are specified. A future evaluation framework addresses functional reliability, posture-feature accuracy, safety-rule behavior, recommendation consistency, and usability. The paper distinguishes the current interface prototype from planned capabilities and makes no claim of clinical validation or demonstrated health benefit.**

***Index Terms*—artificial intelligence, software engineering, decision support, computer vision, pain assessment, personalized exercise, mobile health, safety screening, explainable AI**

I. Introduction

Artificial intelligence (AI) and mobile health tools increasingly support physical assessment, exercise, rehabilitation, and self-management. Computer vision can extract posture-related information from images, while digital applications can organize reported symptoms and suggest activities. A systematic review found low-certainty evidence that mobile applications supporting therapeutic exercise may improve pain and self-reported physical function in musculoskeletal conditions [1]. These findings are promising, but individual tools often address only one part of a user's condition.

A posture application may analyze alignment without considering fatigue or pain, whereas a symptom tracker may record discomfort without explaining an appropriate next step. Evidence from personalized musculoskeletal decision-support studies is also mixed [2], [3]. Recent work on large language models highlights additional risks when exercise guidance is generated without sufficient safeguards [4], [5]. Thus, the challenge is not merely to generate a personalized suggestion but to integrate heterogeneous information within a transparent and safety-aware workflow.

BodyBalance AI is proposed as a mobile framework that combines daily physical and mental check-ins, a structured Pain Locator, user history, and optional BodyScope posture analysis. The aim of this paper is to synthesize relevant research, define the integration problem, present a modular software pipeline, and propose a testable evaluation plan. BodyBalance AI is intended to support daily wellness decisions, not diagnose injuries or replace qualified healthcare professionals.

II. Background and Related Work

A. Computer Vision and Posture Assessment

Image-based pose estimation can provide posture and movement features without specialized motion-capture equipment. A MediaPipe-based physiotherapy study reported 99.22% average classification accuracy for the specific upper-limb movements evaluated, using pose estimation with conventional classifiers [6]. Al Dibs et al. investigated computer vision and deep learning for posture assessment and physiotherapy recommendations [7]. However, task-specific classification accuracy cannot be generalized to all anatomical regions or clinical assessments.

A photographic comparison involving 39 participants found that agreement between ChatGPT and physiotherapists varied by anatomical region, with lower agreement for several shoulder, pelvic, and foot assessments [8]. BodyScope therefore treats image-derived measurements as optional supporting features rather than diagnostic conclusions.

B. Digital Pain Representation

Interactive body maps allow users to record where pain occurs and how it is distributed. Studies have examined the reliability of digitized pain maps [9] and the validity of automated pain-drawing quantification [10].

A scoping review of digital pain manikins identified location, extent, and widespreadness as common measurements, while also noting variation in data representation and analysis [11]. These findings support a structured Pain Locator for self-reported symptoms, without suggesting that a pain drawing establishes the underlying medical cause.

C. Personalized Decision Support and Exercise Guidance

The PERFECT framework combined smartphone and wearable data with personalized physical-activity recommendations. Its proof-of-concept evaluation included 12 participants over 12 weeks, supporting technical feasibility while limiting the generalizability of its findings [12].

Nevertheless, clinical evaluations caution against assuming that personalization alone improves outcomes.

The SupportPrim cluster-randomized trial evaluated personalized clinical decision support in primary physiotherapy care. At 12 weeks, improvement in Global Perceived Effect was reported by 55.4% of the intervention group and 54.8% of the control group, without a clear between-group difference. However, clinically important improvement in patient-specific function favored the control group (59.7% versus 70.3%) [2].

Similarly, a randomized clinical trial of an AI-based musculoskeletal self-management application found no significant improvement in its primary musculoskeletal-health outcome at three months compared with the control conditions [3].

Reviews of therapeutic-exercise applications and software-based low-back-pain interventions offer broader context, but findings depend on intervention design and evidence quality [1], [13].

Research on structured exercise for chronic low-back pain and reviews of practice guidelines also illustrate the importance of tailoring activity decisions to clinical context [14], [15].

D. AI-Generated Recommendations and Safety

Large language models can generate detailed exercise advice, but fluency does not establish correctness or safety. A systematic review of exercise recommendations generated by large language models identified potential benefits alongside concerns about reliability, individualization, and safety [4].

Choi et al. evaluated AI-generated exercise prescriptions for three fictional high-risk clinical scenarios using expert assessments. The study highlighted the importance of professional verification and did not establish clinical safety or effectiveness in real patients [5].

Literature-driven rule generation offers an alternative direction for more constrained therapeutic-exercise logic, although its proposed use does not establish clinical effectiveness [16].

Clinical guidelines provide important context for identifying situations that may require professional assessment. However, a systematic review of neck-pain guidelines found substantial variation and limited agreement regarding recommended red flags [17].

The Saudi clinical practice guideline provides additional evidence-based context for low-back-pain assessment [18].

Neither publication independently validates the eight proposed BodyBalance AI screening indicators.

Explainability and human oversight are therefore relevant design principles [19]. BodyBalance AI proposes explicit screening before routine recommendation generation, rather than treating unrestricted text generation as a safety mechanism.

E. Research Gap

The reviewed work addresses pose estimation, digital pain mapping, exercise personalization, and clinical safety, but these functions are commonly examined separately.

A posture measurement does not describe the user's full daily condition; a pain map does not independently determine safe activity; and a recommendation engine may lack a distinct escalation pathway.

BodyBalance AI addresses this integration gap at the software-architecture level by combining current and historical inputs while separating assessment, safety screening, recommendation selection, and professional-evaluation guidance.

This is a proposed engineering contribution, not evidence of improved clinical outcomes.

III. Problem Definition

People who exercise regularly may need to decide whether to maintain activity, reduce intensity, focus on recovery, or seek advice when they experience fatigue, poor sleep, stress, or pain.

Relevant information can be fragmented across activity trackers, symptom records, and posture-assessment tools.

Four limitations motivate the proposed system:

(1) Check-in information is disconnected from pain and activity history;

(2) Single assessment methods provide incomplete context;

(3) Automated guidance may overlook concerning indicators;

(4) Earlier symptoms and recommendations may not be organized for monitoring or authorized professional review.

BodyBalance AI is designed around four input groups: daily physical and mental check-ins; pain location and intensity; recent activity and user history; and optional image-based posture features.

Its six predefined output categories are maintain activity, modify activity, reduce intensity, focus on recovery, monitor, and consider professional evaluation.

The system is intended for non-diagnostic physical-wellness decision support; it is not designed to diagnose disease, rule out injury, or prescribe treatment.

IV. Technical Methodology

The proposed methodology separates user input, optional posture-feature extraction, safety screening, recommendation selection, and explanation into independently testable components.

This modular design permits operation without an image and makes safety behavior explicit.

The architecture and its planned evaluation are described below.

A. BodyBalance AI Pipeline

The pipeline transforms available self-reported information and optional posture features into a next-step category.

Safety screening occurs before routine recommendation selection.

A predefined concerning indicator diverts the process to professional-evaluation guidance rather than routine activity advice.

Figure 1 shows the proposed components and the two decision paths.

1) Input Collection and User Profile: The proposed input layer organizes daily check-in responses on energy, fatigue, sleep, stress, recent activity, and reported symptoms.

The Pain Locator is specified to let users select up to three body regions and record intensity, onset, reported cause, and change over time.

Research on digital body maps supports structured recording of pain location and extent [9]–[11], but these reports cannot identify the medical cause.

The design also allows relevant previous check-ins, pain entries, and recommendations to provide historical context.

Input validation and missing-data handling should occur before further processing; this stage does not require generative AI.

2) Optional BodyScope Assessment: BodyScope is a planned optional computer-vision component.

A user may capture or upload a posture image, subject to image-format, size, and quality requirements.

The proposed implementation would extract pose landmarks and derive features such as shoulder inclination and pelvic tilt.

The requirements specify a target of 33 landmarks; this is a design target, not a demonstrated extraction result.

Landmark-based posture studies motivate this approach [6], [7], while differences in agreement across anatomical regions require caution [8].

If no valid image is available, the pipeline must continue using the structured inputs without inventing posture measurements.

BodyScope features are supporting observations, not diagnoses.

3) Safety Screening and Decision Logic: A distinct screening component is planned to evaluate predefined concerning indicators before routine activity guidance.

The requirements describe eight red-flag indicators, but their clinical appropriateness and implementation must be evaluated before deployment.

If a defined condition is triggered, the system should suppress routine recommendations and present professional-evaluation guidance.

This behavior is a software safety rule, not a diagnosis or a guarantee that other conditions are safe.

Clinical guidelines and red-flag reviews motivate expert review of the rules [17], [18].

These publications provide clinical context but do not validate the eight proposed BodyBalance AI screening indicators.

For cases not diverted by screening, recommendation logic is designed to combine available check-in, pain, activity, history, and optional posture information and select one of the six categories defined in Section III.

The proposed logic should remain auditable and constrained.

Context-aware personalization research motivates the approach [12], while reviews of LLM exercise advice support avoiding unrestricted generative output as the primary safety mechanism [4], [5].

4) Output, Explanation, and Professional Escalation: The planned output presents the selected category with an explanation tied to information actually provided or processed, such as fatigue, pain intensity, recent activity, or an optional posture feature.

A triggered safety rule instead leads to professional-evaluation guidance.

The design anticipates authorized sharing of relevant records with a qualified professional, subject to consent and access controls.

A history of check-ins and recommendations would support later monitoring.

These modules are separated to permit traceability and independent testing; their existence in the architecture does not imply that every component has already been implemented.

Current Development Status.

The project has an initial application-interface prototype, including a daily check-in flow and dashboard.

The complete Pain Locator workflow, BodyScope processing, validated red-flag screening, recommendation engine, historical integration, and professional-sharing pathway are described here as proposed capabilities rather than completed or clinically tested functions.

B. Proposed BodyBalance AI Evaluation

The following evaluation plan is intended for future implementation and testing.

It does not report completed experiments.

Technical accuracy, safety-rule correctness, usability, and clinical effectiveness are distinct outcomes; positive results in one do not establish the others [2], [3].

1) Technical and Functional Evaluation: Functional testing should verify that daily check-ins and Pain Locator entries are validated, saved, retrieved, and displayed correctly.

Suggested measures include input-completion rate, record completeness, and retrieval consistency.

BodyScope testing should use varied images and reference annotations to measure landmark detection, posture-feature error, and performance across anatomical regions [8].

Integration tests should include missing inputs, absent or invalid images, and incomplete history.

These tests assess software behavior, not clinical benefit.

2) Safety and Recommendation Evaluation: A scenario-based test set should include both routine inputs and predefined concerning indicators.

Expected outputs must be specified in advance, and tests should check whether the appropriate rule fires, routine guidance is suppressed when required, and the escalation pathway remains consistent.

Measures can include rule coverage, decision consistency, and missed predefined escalation scenarios.

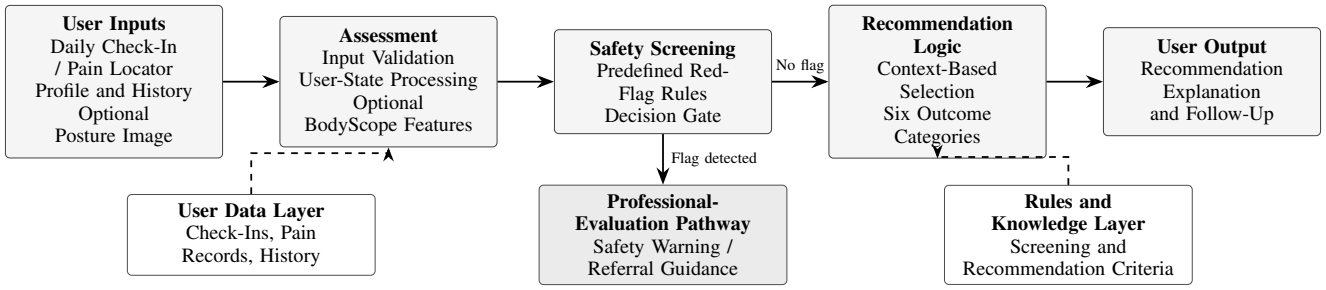


Fig. 1. Proposed BodyBalance AI architecture. The safety gate separates routine recommendations from professional-evaluation guidance. Dashed arrows indicate supporting information sources, not demonstrated production integrations.

Qualified healthcare professionals should review the screening rules and the appropriateness of reference recommendations before any clinical use [17], [18].

For routine scenarios, the selected category can be compared with expert-reviewed expectations.

Generated explanations should be checked for unsupported or overconfident claims, reflecting documented concerns with AI exercise guidance [4], [5].

3) Usability and Explainability Evaluation: Usability testing should assess whether users can complete a check-in, record pain, understand the next-step category, and review available history.

Measures may include task-completion rate, completion time, reported difficulty, and satisfaction.

Explanation testing should check that users understand which inputs influenced the output and when professional assessment is advised.

Any sharing workflow should be tested for permission enforcement and clear user control.

Explainability and human oversight are relevant evaluation principles [19].

Claims about reduced pain, improved recovery, or injury prevention would require separate appropriately designed clinical studies.

V. Discussion and Analysis

The reviewed literature provides evidence for individual technologies used in digital physical assessment, but it also reveals an important distinction between technical performance and meaningful health outcomes. For BodyBalance AI, this distinction influences both the system architecture and the claims that can reasonably be made about its proposed capabilities.

A. Cross-Study Interpretation

Technical accuracy versus real-world applicability. High classification performance in a controlled movement-recognition task does not establish reliable posture assessment across different users, anatomical regions, or image conditions [6], [8]. Consequently, BodyScope should not be treated as an independent decision-maker. Its measurements must remain optional, with the system able to operate when image data are unavailable or unreliable.

Personalization versus demonstrated effectiveness. The reviewed studies represent different levels of evidence. A small proof-of-concept study can demonstrate that personalized recommendations are technically feasible [12], whereas randomized trials evaluate whether such approaches improve patient outcomes. The mixed findings from SupportPrim and AI-assisted self-management research [2], [3] show why feasibility should not be confused with effectiveness. BodyBalance AI therefore defines its outputs as decision-support categories rather than validated recovery prescriptions.

Automation versus safety and accountability. Research on AI-generated exercise recommendations identifies risks when outputs are not sufficiently constrained or professionally reviewed [4], [5]. At the same time, variation between clinical guidelines makes it difficult to assume that a fixed set of warning indicators is universally appropriate [17], [18]. These findings favor a modular design in which safety screening, recommendation selection, and explanation are independently testable and subject to expert review.

Structured information versus clinical interpretation. Digital pain maps support consistent recording of symptom location and distribution, but they do not identify the cause of pain [9]–[11]. Combining pain records with check-ins and activity history may provide a more organized view of a user's reported condition. However, this integration should not be interpreted as an improvement in diagnostic accuracy without further evaluation.

B. Engineering Implications and Limitations

Across these themes, the proposed contribution of BodyBalance AI is architectural integration rather than a new clinical prediction model. Separating data collection, optional computer vision, safety rules, and recommendation logic supports traceability, independent testing, and clearer responsibility for system outputs.

The main trade-off is between providing useful personalized guidance and avoiding unsupported clinical conclusions. Constrained outputs and an explicit escalation pathway may improve the transparency of the design, but their actual reliability remains untested.

The current prototype covers only initial interface functionality. No completed study has established BodyScope measurement accuracy, the validity of the eight screening indicators,

recommendation appropriateness, usability, or improved recovery outcomes for BodyBalance AI. Future evaluation must therefore examine technical correctness and clinical relevance separately, with qualified professional involvement before any health-related deployment.

VI. Conclusion

This paper presented BodyBalance AI as a proposed non-diagnostic mobile decision-support framework integrating daily physical and mental check-ins, a structured Pain Locator, historical context, and optional BodyScope posture features.

Related research supports the technical relevance of these components while highlighting limits in posture assessment, personalization, and AI-generated exercise guidance.

The proposed modular architecture separates assessment from safety screening, routine recommendation selection, explanation, and professional-evaluation guidance.

Six predefined outcome categories and a testable evaluation framework provide a basis for further software development.

The current contribution is a literature-informed design and evaluation plan, supported by an initial interface prototype; it does not establish clinical effectiveness.

Future work should implement and test the planned modules, evaluate posture features across varied conditions, validate safety rules with qualified professionals, and conduct usability and recommendation-quality studies.

Any claims of health benefit would require additional clinical evidence.

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